Evaluation of Measurement Uncertainty

[In accordance with ISO Guide 98-3: 2008]

Expansion and Bleeding of Freshly Mixed Grouts for Preplaced-Aggregates Concrete

ASTM C940:2022

1 **Test equipment**

Glass graduate (1000mL)

Glass graduate (25mL)

2 **Mathematical model**

Expansion % = $\frac{H_3 - H1}{H_1} \times 100$

Bleeding % = $\frac{H_3 - H_2}{H_1} \times 100$

Free Expansion % = $\frac{H_2 - H_1}{H_1} \times 100$

Final bleeding % = $\frac{H_w}{H_1} \times 100$

H_1 = volume of sample at beginning of test, mL,

H_2= volume of sample at prescribed intervals, measured at upper surface of water l

H_3 = volume of grout portion of sample at prescribed intervals, at upper surface of

H_w = volume of decanted bleed water, mL.

3 **Uncertainty equation**

Expansion % = $\frac{H_3 - H1}{H_1} \times 100$ Free Expans

with the sensitivity coefficient: with the sensit

c(H1) = $\partial Ex/\partial_{H1}$= -H3/H1^2 c(H1) = ∂FEx/

c(H3) = $\partial Ex/\partial_{H3}$= 1/H1 c(H2) = ∂FEx/

Bleeding % = $\frac{H_3 - H_2}{H_1} \times 100$ Final bleedi

with the sensitivity coefficient: with the sensit

c(H1) = $\partial BL/\partial_{H1}$= -(H3-H2)/H1^2 c(H1) = ∂FB/∂

c(H2) = $\partial BL/\partial_{H2}$= -1/H1 c(Hw) = ∂FB/

c(H3) = $\partial BL/\partial_{H3}$= 1/H1

4 **Source of uncertainty**

u(H1) : uncertainty of the volume of sample at beginning of test

- calibration uncertainty of the 1-L glass graduate, u(H11)
- uncertainty due to resolution of the glass graduate, u(H12)
- uncertainty due to error of the glass graduate, u(H13)

- uncertainty due to operator, u(H14)

u(H2) : uncertainty of volume of sample at prescribed intervals, measured at upper surface

- calibration uncertainty of the 1-L glass graduate, u(H21)
- uncertainty due to resolution of the glass graduate, u(H22)
- uncertainty due to error of the glass graduate, u(H23)
- uncertainty due to operator, u(H24)

u(H3) : uncertainty of volume of grout portion of sample at prescribed intervals, at upper su

- calibration uncertainty of the 1-L glass graduate, u(H31)
- uncertainty due to resolution of the glass graduate, u(H32)
- uncertainty due to error of the glass graduate, u(H33)
- uncertainty due to operator, u(H34)

u(Hw) : uncertainty of volume of decanted bleed water, mL.

- calibration uncertainty of the 25-mL glass graduate, u(H41)
- uncertainty due to resolution of the glass graduate, u(H42)
- uncertainty due to error of the glass graduate, u(H43)
- uncertainty due to operator in decant of water, u(H44)

5 **Input data**

1-L glass graduate Code:

Calibration uncertainty	(mL)		(k =		)
Resolution	(mL)				
Error	(mL)				
ESDM (3 results) H1	(mL)				
ESDM (3 results) H2	(mL)				
ESDM (3 results) H3	(mL)				

u(Hx1)	u(Hx2)	u(Hx3)	u(Hx4)	u(Hx)	
(mL)	(mL)	(mL)	(mL)	(mL)	
#DIV/0!	0	0	0	#DIV/0!	u(H1)
#DIV/0!	0	0	0	#DIV/0!	u(H2)
#DIV/0!	0	0	0	#DIV/0!	u(H3)

25-mL glass graduate Code:

Calibration uncertainty	(mL)		(k =		)
Resolution	(mL)				
Error	(mL)				
ESDM (3 results)	(mL)				

u(Hw1)	u(Hw2)	u(Hw3)	u(Hw4)	u(Hw)
(mL)	(mL)	(mL)	(mL)	(mL)
#DIV/0!	0	0	0	#DIV/0!

Parameter	a	b	c	SD

H1				#DIV/0!
H2				#DIV/0!
H3				#DIV/0!
H4				#DIV/0!

Results:

H1 (mL)	H2 (mL)	H3 (mL)	Hw (mL)
#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!

6 Evaluation of measurement uncertainty

u(Ex) (Expansion)

Parameter	Factor	a	ci (sensitivity coeff.)	u(xi)*ci	ci*u(xi)^2	γ
Expansion	H1	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	10000
	H3	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	10000
				sum	#DIV/0!	
				uc	#DIV/0!	
				Veff	#DIV/0!	
				t value	2	
				U95	0.0000	

u(FEx) (Free Expansion)

Parameter	Factor	a	ci (sensitivity coeff.)	u(xi)*ci	ci*u(xi)^2	γ
Expansion	H1	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	10000
	H2	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	10000
				sum	#DIV/0!	
				uc	#DIV/0!	
				Veff	#DIV/0!	
				t value	2	
				U95	0.0000	

u(BL) (Free Expansion)

Parameter	Factor	a	ci (sensitivity coeff.)	u(xi)*ci	ci*u(xi)^2	γ
Expansion	H1	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	10000
	H2	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	10000
	H3	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	10000

sum	#DIV/0!	
uc	#DIV/0!	
Veff	#DIV/0!	
t value	2	
U95		0.0000

u(FB) (Final bleeding)

Parameter	Factor	a	ci (sensitivity coeff.)	u(xi)*ci	ci*u(xi)^2	γ
Expansion	H1	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	10000
	Hw	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	10000

sum	#DIV/0!	
uc	#DIV/0!	
Veff	#DIV/0!	
t value	2	
U95		0.0000

ayer, mL,
grout, mL,

sion % = $\frac{H_2 - H_1}{H_1} \times 100$

ivity coefficient:
∂_{H1}= -H2/H1^2
∂_{H2}= 1/H1

ng % = $\frac{H_w}{H_1} \times 100$

ivity coefficient:
∂_{H1}= -Hw/H1^2
∂_{Hw}= 1/H1

ci*u(xi)^4/γ
#DIV/0!
#DIV/0!

#DIV/0!

mL

ci*u(xi)^4/γ
#DIV/0!
#DIV/0!

#DIV/0!

mL

ci*u(xi)^4/γ
#DIV/0!
#DIV/0!
#DIV/0!

#DIV/0!

mL

ci*u(xi)^4/γ
#DIV/0!
#DIV/0!

#DIV/0!

mL